ANKITA SAMADDAR

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EDUCATION

Ph.D. in Computer Science and Engineering, Nanyang Technological University (NTU), Singapore
Thesis : Schedule Randomization based Countermeasures Against Timing Attacks in Real-Time Wireless Networks Supervisor : Arvind Easwaran Co-Supervisor : Rui Tan
CGPA : 5.0/5.0 Aug 2017 - Apr 2022

Master of Technology (M.Tech) in Computer Science, Indian Statistical Institute (ISI), Kolkata
Percentage : 81% with First Class Honours Jul 2015 - Jul 2017

Bachelor of Engineering (B.E.) in Computer Science and Technology, Indian Institute of Engineering Science and Technology (IIEST), Shibpur
CGPA : 8.47/10.00 with First Class Honours Jul 2011 - May 2015

AWARDS AND ACCOMPLISHMENTS

1. ***Best Paper Award in IEEE International Conference on AI in CyberSecurity (ICAIC), 2025*** for "Out-of-Distribution Detection for Neurosymbolic Autonomous Cyber Agents".
2. ***First Runner Up in "Graduate Research Symposium, 2019"***, from School of Computer Science & Engineering, NTU for Ph.D. research presentation.
3. ***Finalist in robotics competition in "International Symposium on Embedded computing and system Design (ISED) 2016"*** at IIT Patna, India, for designing an IoT-enabled warehouse.
4. ***Govt. of India scholarship (Ministry of Human Resource Development, Department of Higher Education), 2010***, for being within top 20 out of 4 million candidates in Higher Secondary Examination.

EXPERIENCE

Vanderbilt University : *Postdoctoral Fellow* Feb 2024 - present
Supervisor : Xenofon Koutsoukos, Sandeep Neema (working on NSA & DARPA projects on cybersecurity)
1. Designing Reinforcement Learning based Neurosymbolic autonomous cyber-agents for enterprise networks.
2. Evaluate the agents by incorporating them into MITRE ATT&CK like environment such as CybORG CAGE Challenge 2 and a private enterprise network testbed at Vanderbilt University.
3. Provide runtime safety assurance of autonomous networks by detecting anomaly or out-of-distribution (OOD) situations using machine learning techniques, e.g., Probabilistic Neural Networks.
4. Vulnerability analysis using imitation learning, e.g., behavior cloning to learn red agent policies from observations in partially observable environments using multi-stage Deep Neural Networks.
5. Design LLM based interpretable cyber-defense agents for enterprise networks.

Tools & Testbed: Private enterprise network testbed at Vanderbilt, CybORG CAGE Challenge 2, Behavior Trees, Blackboards, PyTrees, PyTorch, Pandas, Neural Networks, Supervised Learning.
Proposal: Proposal on "AI-enabled Smart Waste Auditing" using AI, Computer Vision, Image Processing, and Generative AI got shortlisted in the first round by Nashville Innovation Alliance's Civic Tech Jam (final outcome due in Summer 2026).

Continental-NTU Corporate Lab, Singapore : *Research Fellow* Aug 2022 - Sep 2023
Worked in collaboration with BlueSG, an electric car sharing service provider in Singapore
1. Identified vulnerabilities and threats for different applications in Electric Vehicles.
2. Identified cyber-attacks (GPS spoofing attacks) in IEEE 802.11p based V2X communications and its consequences in Vehicle Rebalancing application.

3. Proposed and implemented a location estimation-based attack detection technique and a MAC authentication technique as a countermeasure to defend the system and prevent its misuse.
Tools & Testbed: Traffic Control Interface (TraCI) and SUMO simulator.

PROJECTS

Nanyang Technological University, Singapore : *PhD Research* Aug 2017 - Apr 2022
Thesis : Schedule Randomization based Countermeasures Against Timing Attacks in Real-Time Wireless Networks
Supervisor : Arvind Easwaran, Co Supervisor : Rui Tan;
Worked in collaboration with Delta Electronics Inc., Delta-NTU Corporate Laboratory for Cyber-Physical Systems, National Research Foundation (NRF), Singapore
1. Proposed threat models for timing attacks (e.g., selective jamming attacks) in time-critical and safety-critical cyber-physical systems, e.g., Industrial Control Systems (ICS), leading to denial of service (DoS) in specific time-critical applications resulting in catastrophic consequences to the system.
2. Proposed schedule randomization based countermeasures, (centralized and distributed strategies) against timing attacks for time-critical wireless networks in ICS, e.g., IEEE 802.15.4 based WirelessHART networks, and Ultra Reliable Low Latency Communication (URLLC) in 5G networks.
3. Integrated Period Adaptation while ensuring system performance to increase the degree of randomization and to reduce energy consumption of the system.
4. Proposed to use Kullback-Leibler divergence and Prediction Probability as security metrics to evaluate the performance of the proposed countermeasures.
Tools & Testbed : Private wireless sensor network testbed in National University of Singapore. IEEE 802.15.4 compliant CC2420 devices, GISOO simulator, Cooja simulator, TinyOS, TelosB and TmoteSky motes, Simulink.

Indian Statistical Institute : *Final Year M.Tech dissertation* Dec 2016 - June 2017
Mentor : Ansuman Banerjee
1. Proposed a novel, bank-aware memory scheduling policy for DRAM controllers to handle real-time memory requests with hard deadlines for mixed criticality systems.
2. Modified the source code of DRAM simulator, "Ramulator", to support real-time memory requests and compared the approach with the SOTA DRAM scheduling policies.

Bosch India : *Project* Jan 2016 - March 2016
Mentor: Ansuman Banerjee, Dakshina Dasari
1. Worked in a project on task partitioning and scheduling in real-time multicore systems for autonomous vehicles.

IIEST, Shibpur : *Final Year B.E. dissertation* Jul 2014 - May 2015
Mentor : Somnath Pal
1. Proposed an efficient generalized algorithm that can generate Bond Electron (BE) matrix for any chemical reaction. Extended the existing BE matrix to represent all types of possible chemical reactions.
2. Proposed an automated template generation software for mining chemical graphs from the BE matrix.
3. Implemented all techniques in Python.

INTERNSHIPS

Nanyang Technological University, Singapore : *Intern* May 2016 - Jul 2016
Mentor : Arvind Easwaran
1. Formally verified safe functioning of an artificial pancreas in patients with type-1 diabetes. Modeled the functionality of artificial pancreas using hybrid automata.
2. Formally verified safe functioning of the nonlinear hybrid model in dReach, linearized the hybrid model using Jacobian linearization, formally verified the linearized model in SAL with some approximation.
Tools : dReach, SAL.

Indian Statistical Institute : *Intern* June 2014 - Jul 2014
Mentor : B. Uma Shankar

1. Explored different image processing techniques, e.g., histogram analysis, image thresholding, segmentation, etc.
2. Implemented entropy based image segmentation techniques, image thresholding based on clustering, Otsu's method for thresholding in Matlab.

TEACHING EXPERIENCE

1. **Instructor for Cybersecurity course (CS4277) along with Prof. Sandeep Neema at Vanderbilt University (Spring Semester, 2026):**
- Teaching and designing lecture materials and assignments for Cybersecurity course at Vanderbilt University.
2. **Mentor undergraduate students for final year project at Vanderbilt University (Fall Semester, 2025, Spring Semester, 2026):**
- Supervising a Final Year undergraduate student on a project on designing LLM-based cyber-defender for enterprise networks.
3. **Instructor for Competitive Coding, NTU (Fall Semester 2018):**
- Teaching different advanced data structures (Trie, Hastables, Double Linked Lists, etc) and algorithms, and their implementation for competitive coding.
- Hosted practice coding sessions and discussed efficient solutions.
4. **Mentored undergraduate students for their project at NTU (Fall Semester 2019, Spring Semester 2020, Fall Semester 2020):**
- Supervised Final Year undergraduate students for implementing AES-128, AES-192 and AES-256 to prevent network packets from integrity attacks on a Fischertechnik Sorting Line with Color Detection as part of their Final Year undergraduate project.
- Supervised Final Year undergraduate students to implement Process Scheduler and Memory Management Unit in OS161 as part of their Final Year undergraduate project.
5. **Teaching Assistant of Engineering Mathematics I (CY1601/RE1001) and II (CY1602/RE1021), SCSE, NTU (Fall Semester 2017, Spring Semester 2018):**
- Lab Instructor; Introduced basic engineering mathematical functions in Matlab and Scilab.
- Assisted students to implement complex mathematical functions using basic engineering mathematical functions.
6. **Teaching Assistant for Operating Systems (CE2005/CZ2005), SCSE, NTU (Fall Semester 2018):**
- Lab Instructor; Designed assignments on integration of scheduler (First Come First Serve and Round Robin) in Nachos operating system.
- Instructed students to implement process scheduler, process synchronization, page table and memory management in Nachos operating system.
- Graded assignments and oral examination.
7. **Teaching Assistant for Time Critical Computing (CE4057/CZ4057), SCSE, NTU (Fall Semester 2018):**
- Lab Instructor; Instructed students to implement Rate Monotonic and Earliest Deadline First scheduling policies using recursive data structures in Micrium operating system on a TI Stellaris Evalbot.
- Instructed students to implement Priority Ceiling Protocol and Stack Resource Policy in Micrium.
- Graded course projects.
8. **Teaching Assistant for RTOS for Cyber-Physical Systems (CE7452), SCSE, NTU (Fall Semester 2019):**
- Assisted the post-graduate students in their course projects.
- Graded survey reports and assignments.
9. **Teaching Assistant for Computer Architecture Course (CE3001/CZ3001), SCSE, NTU (Spring Semester 2019):**
- Lab Instructor; Instructed students to implement instruction pipeline architecture in Xilinx ISE Design Suite as part of their coursework labs.
- Graded assignments.

TECHNICAL SKILLS

1. Programming : C, Python, C++, Java, PHP, Javascript, UML.
2. Machine Learning : PyTorch, PyTrees, Numpy, Pandas, Blackboards, Textgrad optimizer with LLM, Prompt Engineering, Zero shot and Few shot learning, Neural Networks, Behavior Trees.
3. Operating Systems: Windows, Linux, FreeRTOS, Micrium.
4. System Modeling: Matlab and Simulink.
5. Real Testbed: Indriya testbed (WSN testbed in NUS), private enterprise network testbed in Vanderbilt.
6. Network Protocols: IEEE 802.15.4 (WirelessHART), 3GPP, URLLC in 5G, IEEE 802.11p (VANET). Familiar with other wireless protocols (ZigBEE, 6LowPAN, etc.) and wired protocols (TTEthernet, Bluetooth, CAN, etc.).
7. Network Simulator: Cooja, TinyOS, CybORG CAGE Challenge 2.
8. Vehicle simulator: Simulation of Urban MObility (SUMO).
9. Verification Tools: UPAAL, dReach, SAL.
10. Hardware platforms: Zedboard, TI Stellaris Evalbot, Arduino Uno, CC2420 devices, Raspberry Pi 3.
11. Hardware Description Language: Verilog.
12. Hardware Design Tools: Xilinx ISE.
13. Miscellaneous: SQL, Elastic Search.
14. Courses: Artificial Intelligence, Distributed Systems, Real-Time Operating Systems, Electronic Design Automation, Pattern Recognition, Data Mining, Computer Vision, Natural Language Processing.

PUBLICATION LIST

1. **Ankita Samaddar**, Sandeep Neema, Daniel Balasubhramanium, Xenofon Koutsoukos, "Learning Red Agent Policy from Observations for Neurosymbolic Autonomous Cyber Agents", IEEE International Conference on Omni-Layer Intelligent Systems (COINS), 2026 (under review).
2. Nicholas Potteiger, **Ankita Samaddar**, Taylor T Johnson, Xenofon Koutsoukos, "Reward Shaping and Action Masking for Compositional Tasks using Behavior Trees and LLMs", IEEE International Conference on Neurosymbolic Systems (NeuS), 2026.
3. Chris Hicks, Elizabeth Bates, Shae McFadden, Isaac Symes Thompson, Myles Foley, Ed Chapman, Nickolas Espinosa Dice, **Ankita Samaddar**, Joshua Sylvester, Himanshu Neema, Nicholas Butts, Nate Foster, Ahmad Ridley, Zoe M, Paul Jones, "Building Better Environments for Autonomous Cyber Defence", AI for Critical National Infrastructure (AI4CNI), AAMAS 2026.
4. **Ankita Samaddar**, Nicholas Potteiger, Xenofon Koutsoukos, "Out-of-Distribution Detection for Neurosymbolic Autonomous Cyber Agents", IEEE International Conference on AI in Cybersecurity (ICAIC), 2025 (***Best Paper Award***).
5. Nicholas Potteiger, **Ankita Samaddar**, Hunter Bergstrom, Xenofon Koutsoukos, "Designing Robust Cyber-Defense Agents with Evolving Behavior Trees", IEEE International Conference on Assured Autonomy (ICAA), 2024.
6. **Ankita Samaddar**, Arvind Easwaran, "A Location Validation Technique to Mitigate GPS Spoofing Attacks in IEEE 802.11p based Fleet Operator's Network of Electric Vehicles", IEEE International Conference on Intelligent Transportation Systems (ITSC), 2024.
7. **Ankita Samaddar**, Arvind Easwaran, "Online Distributed Schedule Randomization to Mitigate Timing Attacks in Industrial Control Systems", ACM Transactions on Embedded Computing Systems (TECS), Volume 22, Issue 6, Article 100, Pages 1-39, November 2023.
8. **Ankita Samaddar**, Arvind Easwaran, "Online Schedule Randomization to Mitigate Timing Attacks in 5G Periodic URLLC Communication", ACM Transactions on Sensor Networks (TOSN), Volume 19, Issue 4, Article 93, Pages 1-26, July 2023.
9. **Ankita Samaddar**, Arvind Easwaran and Rui Tan, "Work Already Published: A Schedule Randomization Policy to Mitigate Timing Attacks in WirelessHART Networks", Brief-Presentations Session of IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS), 2021.
10. **Ankita Samaddar**, Arvind Easwaran, Rui Tan, "A Schedule Randomization Policy to Mitigate Timing Attacks in WirelessHART Networks", Springer Real-Time Systems, Volume 56, Pages 452-489, October 2020.
11. **Ankita Samaddar**, Arvind Easwaran, Rui Tan, "SlotSwapper: A schedule randomization protocol for real-time WirelessHART networks", Real-Time Networks (RTN), 2019 (ACM SIGBED).
12. **Ankita Samaddar**, Zahra RahimiNasab Reza, Arvind Easwaran, Ansuman Banerjee, Xue Bai, "Lineariza-

tion based Safety Verification of a Glucose Control Protocol", IEEE International Symposium on Real-Time Computing (ISORC), 2019.
13. **Ankita Samaddar**, Moumita Das, Ansuman Banerjee, "A new memory scheduling policy for real time systems", International Symposium on Embedded Computing and System Design (ISED), 2017.
14. Shreya Ghosh, **Ankita Samaddar**, Trishita Goswami, Somnath Pal, "Chemical Graph Mining for Classification of Chemical Reactions", International Conference on Computational Intelligence, Communications, and Business Analytics (CICBA), 2017.
15. **Ankita Samaddar**, Trishita Goswami, Shreya Ghosh, Somnath Pal, "An algorithm to input and store wider classes of chemical reactions for mining chemical graphs", IEEE International Advance Computing Conference (IACC), 2015.

Book Chapter: Arvind Easwaran, Michael Yuhas, Saravanan Ramanathan, **Ankita Samaddar**, "Real-Time Scheduling for Computing Architectures", Handbook of Computer Architecture, Springer, 2024.[1]

PROFESSIONAL ACTIVITIES

Editorial Board Member :
International Journal of Wireless Communications and Mobile Computing (Aug 2024 - Aug 2027)
Program Committee Member :
IEEE Real Time Systems Symposium (RTSS), 2021 [Brief Presentation Track
IEEE International Conference on Computer Design (ICCD), 2025
ACM Internet Measurement Conference (IMC), 2025
IEEE Intelligent Vehicles Symposium (IV) 2026
IEEE/ACIS International Conference on Computer and Information Science (ICCIS) 2026
Research Paper Reviewer :
ACM Transactions on Design Automation of Electronic Systems (TODAES)
ACM Transactions on Cyber-Physical Systems (TCPS)
ACM Transactions on Sensor Networks (TOSN)
IEEE Transactions on Signal and Information Processing over Networks (TSIPN)
IEEE International Transportation System Conference (ITSC) 2024, 2025, 2026
International Conference on Big Data Analytics in Bioinformatics (DABCon), 2024
Journal of Supercomputing, Science Publishing Group.

EXTRA-CURRICULAR ACTIVITIES

1. Participated in robotics competition "Kshitij 2013" at IIT Kharagpur, India, in an image processing event "Seekar".
2. Volunteered in ASEAN University Games, 2016 in Singapore.
3. Treasurer of NTU SCSE Graduate Student Club (GSC), 2020-2021.
4. Director of Publicity of NTU SCSE GSC, 2019-2020.
5. Director of Career and Development of NTU SCSE GSC, 2018-2019.
6. Serving as organizing committee member of Vanderbilt Postdoctoral Association Symposium 2025.

[1] All Authors had equal contributions, naming sequence based on sections in the chapter